Self-Fullfilling Prophecy?: Do Subjective Mortality Expectations Matter For Predictive Mortality and Longevity

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Abstract

Subjective expectations about health are thought to matter theoretically for mortality outcomes because they contain rich, but unobserved information about health status. While subjective expectations are correlated with mortality outcomes, whether they enrich the prediction of individual longevity remains less clear. This paper examines whether subjective expectations improve the predictability of longevity compared to sociodemographic and economic covariates using representative panel data and a variety of supervised learning models. Results indicate that subjective expectations contribute little predictive power to models predicting death age, but improve predictions of survival past the age of 75. These results suggest that health surveys may be enriched by incorporating subjective mortality questions.

1 Introduction

The extent to which we can predict mortality well matters for understanding population health and policy decisions to meet the growth of the aging population in the U.S. Recent demographic work has thus aimed to address this problem by leveraging various demographic, forecasting, and prediction methods to study the predictability of longevity. However, recent work shows that individual-level mortality and longevity prediction is challenging, with uncertainty remaining high, even with big microdata and rich individual-level covariates (Breen and Seltzer 2023). Moreover, even in cases where predictive performance is better (i.e., with survival curves), there are notable inequalities in predictive performance across group (Badolato et al. 2023).

Despite empirical challenges, obtaining better predictions of individual-mortality and longevity are crucial for policy and scholarly reasons. First, predicting mortality from individual-level data is important for forecasting population longevity in the future, especially by social groups or categories that are not readily available from tabular data. Generating accurate predictions can inform public health policy decisions allocating resources to populations ar older ages based on predicted life expectancy with morbidities, disabilities, disease, loneliness, and other chronic conditions (Raymo and Wang 2022). Second, because research documents notable inequalities in the predictability of mortality for different social groups, the incorporation of subjective measures may improve predictive accuracy and thereby contribute to efforts to narrow mortality and longevity inequalities (e.g., by race).

One potential avenue for improving predictions, is to leverage datasets, such as the Health and Retirement Study (HRS), that contain rich covariates as well as objective (e.g., functional limitations) and subjective health information. Past work has employed subjective mortality expectation questions in the HRS to aid in interpreting the predictive performance of models, but has overlooked their role as focal features. While current evidence suggests that individuals are not very good at predicting their own mortality and there are noted inequalities in performance by social group, whether subjective probabilities can be used in a prediction model as "utility features" (Kose et al. n.d.) in combination with strongly predictive variables (e.g., education, income, age, health) to achieve gains in predictive performance is currently unexplored.

The present study builds upon past efforts by testing the predictive performance of subjective mortality measures to (1) predict individual longevity (death age in years) and (2), as a secondary exercise, predict survival past threshold ages (e.g., 75 years) relative to models without expectations. To examine the predictability of mortality and longevity, I leverage HRS panel data and a variety of machine learning models to evaluate predictive performance relative to a baseline OLS regression model, predicting longevity and living to 75 years as a function of subjective probabilities only. For each model, I contrast two feature sets: the first is a set of 82 cross-sectional and panel covariates, and the second feature set adds the subjective information to the 82 covariates.[1]

Results from random forest and XGBoost models combined with rich, panel data reveal that subjective measures do little to improve the predictability of longevity over other demographic, economic, and health covariates. However, subjective expectations do improve the predictability of surviving to age 75 compared to covariates. Comparisons in the specific predictive performance algorithms and variable importance are discussed.

2 Background

The population in the United States and across the world is aging rapidly as human lifespans continue to extend well beyond previously thought limits. A recent report by the Pew Research Center (2024) projected that the population of adults 65 years and older will quadruple by 2054. An increasingly old population poses

[1] Note, the 82 covariates refer to variables in their "unprocessed" form. That is, before creating factors, deciles, and nonlinear transformations of the variables.

many problems related to caring for the old and ensuring enough labor supply to sustain economic productivity. Accordingly, questions related to how long we will live and what factors predict longer and shorter lifespans have become increasingly salient to policymakers, public health researchers, and demographers. An emergent literature in demography has reformulated these questions as a prediction problems that can tackled with sophisticated machine learning methods that leverage advancements in computational capacity (Badolato et al. 2023; Breen and Seltzer 2023).

Using a variety of datasets ranging from the full count census to the Health and Retirement Study (HRS), evidence demonstrates that predicting longevity is challenging, even with large datasets and rich predictors. Despite a wealth of social science evidence on the determinants of mortality, obtaining good predictions across diverse populations is challenging. Breen and Seltzer (2023) show that in both cohort and period analyses predicting the probability that an individual will die in the next ten years, the R^3 for preferred prediction models is low, with as much as 98.6% of the variance unexplained. Using the HRS, Badolato et al. (2023) found better predictive improvements when modeling Cox survival curves, which partly arises from the rich covariate set and longitudinal nature of the data. However, they document stark inequalities in predictability by social group, particularly for race, gender, and education.

A common takeaway of both papers is that mortality and longevity can be unpredictable and prediction quality varies. First, Sociodemographic features provide weak predictive information, despite substantive and theoretical importance (Breen and Seltzer 2023). Second, researchers need to employ a more expansive set of theoretical predictors guided by theory and practical intuition. This paper attempts to make headway on improving our ability to predict mortality and longevity by considering the potential value of individuals subjective beliefs about their own mortality. While Breen and Seltzer (2023) and Badolato et al. (2023) discuss subjective mortality information, they do so in the context of benchmarking estimates but do not explicitly evaluate their predictive performance.

There are a variety of theoretical and empirical reasons why subjective information may be valuable for predicting longevity. First, decades of sociological and economic research recognizes the theoretically predictive power of individual beliefs on actual outcomes. The concept of the self-fulfilling prophecy, originally coined by Merton (1948), which states that beliefs about future events, whether true or false, can engender those events. In the context of health and longevity, this can be seen intuitively by considering that individuals generally have a good sense of their own health and have knowledge about their own individual circumstances that are unobserved by covariates but may predict mortality. Second, in the context of predicting individual longevity, subjective information about health is relatively low cost to collect. Rather than asking complex health screening questions or collecting expensive biomarkers, asking a simple set of questions about subjective views of mortality is relatively easy and low-cost to implement on health surveys. This is one reason why self-rated health questions are common on surveys: they have been validated in various settings and across groups and are predictive of future mortality.

In research on aging and health, subjective mortality probabilities have been employed before in forecasting settings, yet a full assessment of their predictive utility remains far from settled. For instance, one study previously used subjective measures in the HRS to calculate subjective, cohort-life-table estimates for U.S. men and women that were found to be predictive of revisions to Social Security life table estimates (Perozek 2008). Indeed, evidence suggests that subjective expectations have the potential to contribute meaningful information, yet their contributions to predictive exercises remain unclear.

3 Data

3.1 Data Description

To predict longevity, I leverage the Health and Retirement Study (HRS) – a panel dataset of approximately 32,000 U.S. adults 50 years and older. The HRS contains comprehensive information on health, wealth, stock portfolios, social and personal relationships, and other variables related to aging. The HRS also tracks mortality outcomes of sample participants that form the basis of this study. For this exercise, I focus on the sample of individuals who have died and collapse the long panel dataset into a wide dataset, where each row indexes an individual with a single death age and columns index both time invariant covariates (e.g., race, education) and time varying covariates (e.g., income at each wave). This transformation is also advantageous for bootstrapping and cross-validation, because it ensures that individuals' entire sets of person-observations are sampled into a fold or sample to avoid sampling individuals at different time points into different folds or samples, which could pose issues for prediction. Table 1 describes a set of focal (but not all) covariates in detail before pre-processing.

In addition to sociodemographic, economic, and health features, the HRS includes sets of subjective expectations about retirement planning, financial planning, and health. For mortality, subjective expectation questions ask respondents to rate the probability from 0-100 that they will live beyond a certain age (i.e., 75 and 80 years of age) (Hurd and McGarry 2002; Perozek 2008). Specifically, the question asks: "What do you think are the chances that you will live to be 75 or more?" Using the subjective indicators, I test whether they have predictive power for both longevity and survival past 75 years of age. To do so, I examine how well machine learning models leveraging all non-subjective features (Covariate set 1) perform compared to models that include subjective features (i.e., 75 and 80 year expectations) (Covariate set 2).

Table 1: Descriptive Statistics of Health and Retirement Study Sample

	N	Mean	Min	Max
Age	54846	68.31	50.00	79.00
Live to Age 75	54846	0.25	0.00	1.00
Black	54846	0.15	0.00	1.00
White	54846	0.81	0.00	1.00
Other	54846	0.04	0.00	1.00
Hispanic	54837	0.08	0.00	1.00
Female	54846	0.54	0.00	1.00
Subjective Life Expectancy (75+)	54846	61.15	0.00	100.00
Subjective Life Expectancy (80+)	54844	44.40	0.00	100.00
Less than HS	54846	0.32	0.00	1.00
HS	54846	0.49	0.00	1.00
Some College	54846	0.04	0.00	1.00
College	54846	0.09	0.00	1.00
Advanced Degree	54846	0.06	0.00	1.00
Wealth (Q1)	54846	0.23	0.00	1.00
Wealth (Q2)	54846	0.22	0.00	1.00
Wealth (Q3)	54846	0.20	0.00	1.00
Wealth (Q4)	54846	0.18	0.00	1.00
Wealth (Q5)	54846	0.16	0.00	1.00
Total Wealth ($1,000s)	54846	509.52	-3482.96	69076.20
Total non-Housing Wealth ($1,000s)	54846	332.55	-3482.96	51295.42
Total Stock Value ($1,000s)	54846	73.25	0.00	17015.10
N Children	54846	3.57	1.00	11.00
Mean Days of Contact with Children	51298	950.95	1.00	8760.00
Mean Age of Children	54310	41.18	0.00	76.50
Sum of Functional Limitations	54846	0.43	0.00	6.00
Sum of Chronic Conditions	54846	2.20	0.00	8.00

All dollar values are inflation adjusted to 2022 Dollars per the CPI from the Bureau of Labor Statistics. Number of contact days with child is the number days over a two-year period. Number of observations varies by variable because certain values (e.g., subjective life expectancy) that are only observed for those over a certain age. Numbers of observations refer to person-observations over several HRS waves.

There are a few important features of the subjective probabilities that are worth discussing. First, as noted by Perozek (2008) and others, about 2 percent of the respondents in the HRS provide invalid or impossible responses to subjective mortality questions. For instance, some respondents rate the probability that they will survive to 85 years old greater than the probability that they will survive to 75. Researchers have attributed this to a misunderstanding of the question and in settings where life-tables are estimated, they cannot be used because they are internally inconsistent. However, this is less of a concern in a prediction setting where we are concerned about obtaining precise $\hat{Y}$ values rather than correctly estimated βs or survivor curves. In the cases where individual survival probabilities are internally inconsistent, they will either contribute predictive power or not to the resulting predictions.

4 Data Cleaning and Processing

Variable transformations and data processing are crucial to maximizing predictive power. For instance, applying transformations to continuous that bin values into new levels allows for greater non-parametric or non-linear structure to be revealed, which in many cases improves performance. To capture a richer structure from the available data, I apply a series of feature engineering steps described below:

- Create a new level indicator for factor variables, that have differing levels in the training and testing set.
- Create a new level for factor variables called "unknown" to code them as missing (this will be used as a missing indicator later).
- Impute numeric variables with the mean value of non-missing values and the mode for nominal variables.
- Remove variables that have zero variance.
- Convert all nominal variables to dummy variables.

After the pre-processing steps are applied to the data, the resulting dataset comprises 17.2 thousand observations and 647 predictors. To train the model, I randomly sample 80% of the data for the training sample and 20% for the hold-out sample to evaluate predictions.

5 Methods

5.1 Overview of Prediction Framework

The goal of this paper is to predict longevity (death age in years) among those who have died in the HRS sample with a secondary analysis classifying individuals based on whether they lived to age 75. In this section, I describe the preferred prediction model for the continuous case. To predict longevity, I employ a variety of supervised machine learning algorithms ranging from Ordinary Least Squares (OLS) regression, to tree based methods (trees, forests, boosted trees) to find the algorithm that best maps *death age* as a function of predictors $f(X)$.

For both longevity and survival to age 75, my preferred model is a random forest. Tree-based methods generate predictions by partitioning the predictor space (i.e., Xs) into non-overlapping groups and taking the average value of death age in each group. The decision to split on a variable is guided by maximizing a goodness of fit statistic (in this case RMSE (described in **Asessing Goodness of Fit**) and is local to each specific step in the decision tree. That is, the model builds trees via the splitting objective to achieve the optimal improvement in goodness of fit at each potential split. This is called recursive, binary splitting; Rather than run a costly optimization routine throughout an entire tree, the algorithm works in a stepwise fashion. The random forest model builds on the simple regression tree in a few important ways. First, random forests comprise multiple trees (hence the forest) that are generated from bootstrap samples of the training data that aide in reducing the variance in predictions from a single tree. Second, at each split, random forests only consider a random subset of predictors to split on, which helps de-correlate trees and avoid the problem of the algorithm always selecting "highly predictive" features to split on and thus undermining the advantage of averaging predictions across multiple trees.

To generate predictions, I train a random forest model using 100 trees and use five-fold cross-validation to tune the number of covariates sampled for each split and the minimum number of nodes. For the regression, I use RMSE as the loss function for death age prediction and the log-loss for survival classification. The chosen values of the tuning parameters for the main models predicting death age are reported in Table 4 in the **Appendix**.

5.2 Assessing Goodness of Fit

To assess goodness of fit, I use a variety of measures for each outcome. For death age, I use the RMSE and R^2. The two goodness of fit measures are calculated as follows.First, RMSE is calculated as.

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{Y}_i - Y_i)^2} \tag{1}$$

The RMSE measures the average of the deviations between predicted and actual values. The RMSE can be naturally interpreted against the standard deviation of the outcome variable (death age) as how well the model does relative to simply using the average value to predict the outcome. A lower RMSE value indicates better predictive performance.

Second, I use the R^2, which is calculated as follows.

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(Y_i - \hat{Y}_i)^2}{\sum_{i=1}^{N}(Y_i - \bar{Y})^2} \tag{2}$$

The R^2 ranges from $-\infty$ to 1 and measures the share of the variance explained by the model. A higher R^2 value indicates better predictive performance.

For predicting survival past 75, I use the accuracy, sensitivity, precision, and area under the curve (AUC). Each of these measures targets a different aspect of model performance for binary outcomes. The AUC is the focal and most holistic measure that captures the area under the receiver-operator curve (ROC), which gives an overall assessment of the model's ability to distinguish between classifications (i.e., survival to 75 versus dying before 75) rather than the performance of classifying individuals into a certain category.

The accuracy of the model is the proportion of classifications, whether positive or negative, that were correct. The precision is the proportion of those that survived to 75 that were classified correctly out of true and false positives. Finally, the sensitivity, or true positive rate, refers to the proportion of the sample that survived past 75 classified as such out of the total number of individuals who survived past 75.

5.3 Interpreting Models

A key limitation of machine learning methods is that they are often "black boxes" where a researcher specifies a model and obtains predictions, but which features contributed to those outcomes are either not known or usually ignored. I address model interpretability in two ways.

The first is built in to my prediction exercise by design. I contrast two feature sets in all models. One contains 647 covariates spanning health, education, wealth, income, social relationship quality, and others captured in the HRS. The second contains all of the 647 variables as well as a set of subjective health and mortality expectations variables. Therefore, comparing the predictions from these two feature sets both within and across models allows for an examination of variable importance.

To further assess variable importance, I obtain highly important features from random forest models and discuss which features contribute strongly to predictions obtained across models.

6 Results

6.1 Do Subjective Mortality Expectations Matter for Predicting Longevity?

I first present analyses of the absolute and relative performance of different machine learning methods for predicting longevity. Table 3 compares the performance of each method by covariate set, relative to a baseline OLS regression of longevity on mortality expectations. Figure 1 displays results from Table 2 visually. First, a baseline regression of longevity on mortality probabilities has an r-squared of 0.036 indicating that the probabilities explain little of the variance in age at death. The out of sample RMSE for this model is 9.116, which indicates that the model improves only slightly over using the mean value of death age (RMSE = 9.283).

In comparison to the baseline model, adding covariates to OLS improves the RMSE by 36.992 percent. Adding subjective probabilities in addition to covariates improves the the RMSE by 37.065, percent over baseline. However, this relative improvement is small compared to the covariate only model.

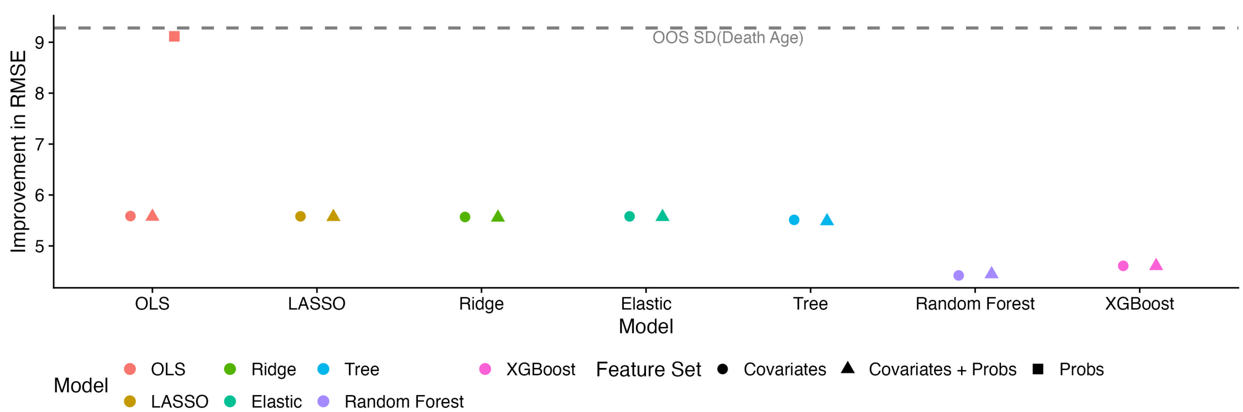


Figure 1: Absolute Improvement in Longevity Prediction

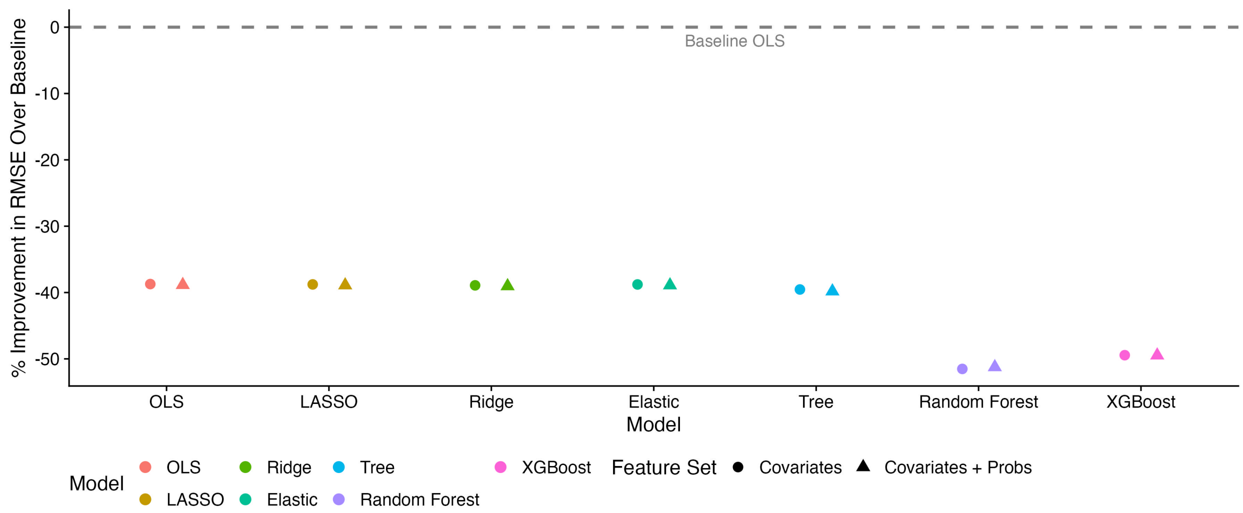


Figure 2: Relative Improvement in Longevity Prediction

Predictive performance is similar with regularized models (LASSO, Ridge, and Elastic Net) and single tree methods, both across covariate sets and models. However, the largest increases in predictive performance are obtained by using tree-based methods such as random forest and XGBoost. In terms of predictive performance, the preferred model is a random forest.

For the model with covariates only, the random forest improves substantially on the baseline RMSE, by 47.962 percent and achieves an R^2 of 0.739. Adding subjective probabilities provides essentially no improvement in predictive power. However, in the XGBoost model, despite performing worse overall than random forest, there are small improvements.

Across quintiles of death age, the random forest model improves on prediction for all quintiles. However, there are notable inequalities in the magnitude of these improvements. The majority of the increase in improvement is in the 3rd quintile, while the lowest improvement is consistently in the 2nd quintile. Overall, the random forest model with covariates and probabilities performs the best, but it is not distinguishable from the covariate only random forest.

Next, I describe the secondary prediction exercise that seeks to predict survival past 75 years of age using covariates and subjective probabilities. Across models, subjective probabilities significantly improve on the predictability of survival past 75 years old in all models. Similarly, tree-based methods vastly outperform the traditional regression and regularization methods for classifying individuals. The preferred model for survival is an XGBoost model.

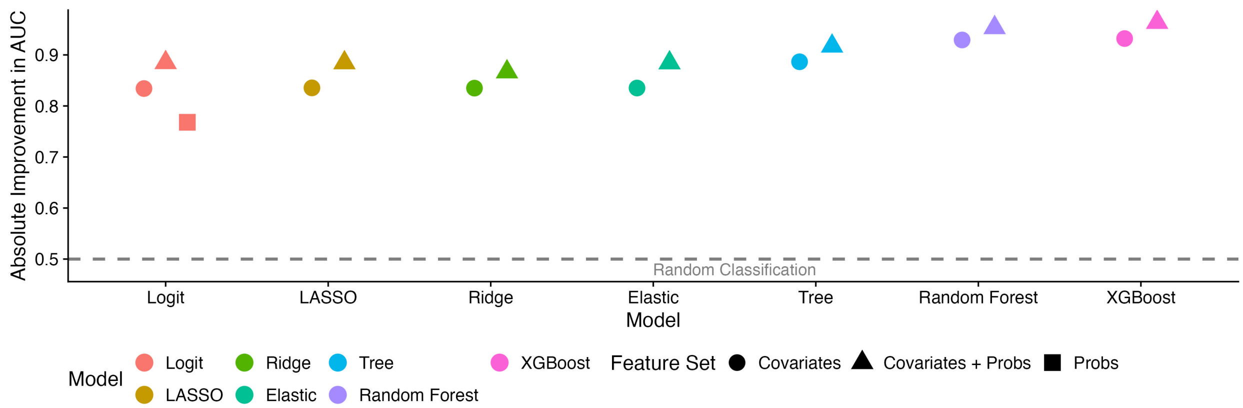


Figure 3: Absolute Improvement in Survival Threshold Prediction

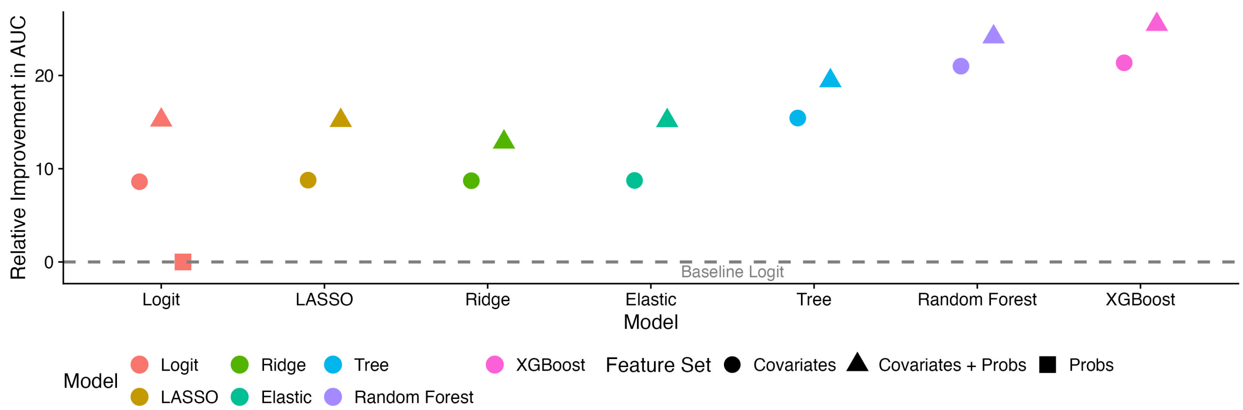


Figure 4: Relative Improvement in Survival Threshold Prediction

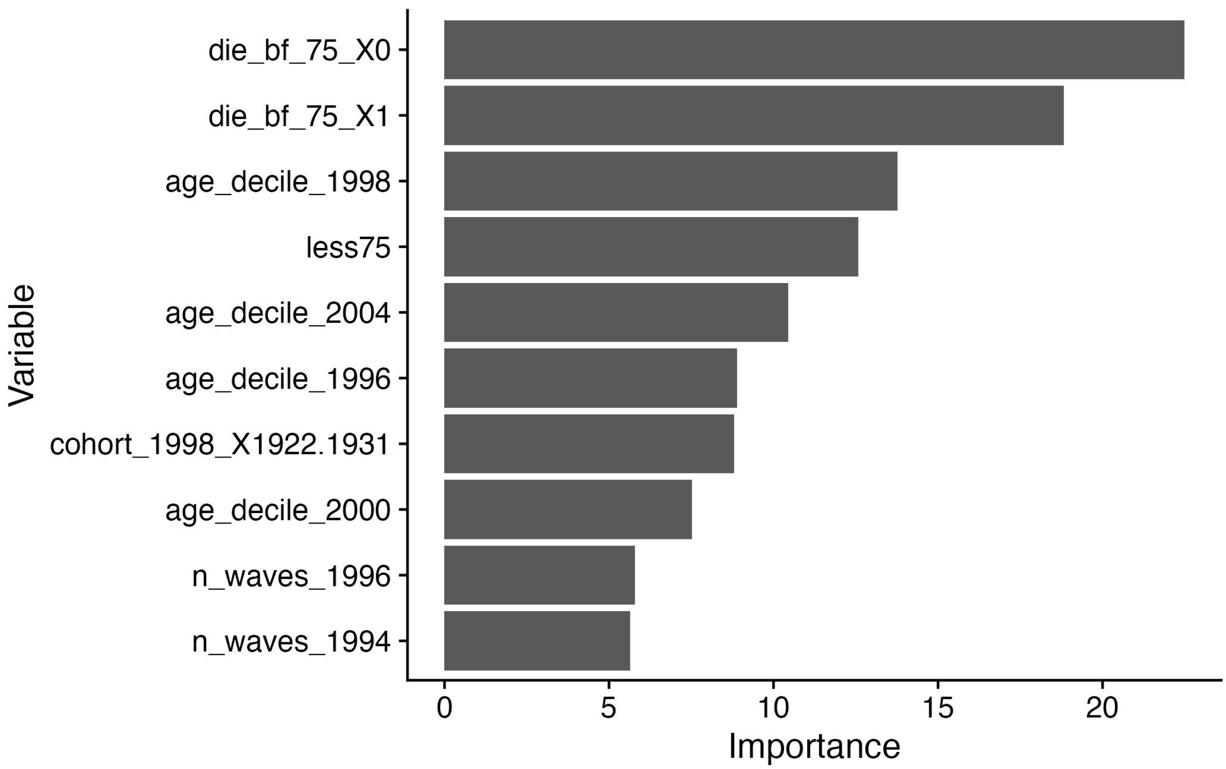
Using a logit model with subjective probabilities achieves a 0.768 AUC, indicating that the model does better at classifying survival relative to a random model that would have an AUC of 0.5. Adding covariates and covariates and probabilities improves the AUC by 8.606 and 15.217 percent, respectively. This suggests that even relative to the naive, baseline model and covariates, the probabilities improve prediction.

Similar to predicting death age, the regularized and single tree models essentially perform the same as the LOGIT specification. The preferred model, based on predictive performance, is the XGBoost model. This model improves predictive performance by 21.359 percent over baseline for the covariate only model and 25.477 percent for the model including probabilities. In both absolute and relative terms, subjective probabilities improve predictive power by about 4 percentage points. In contrast to predicting death age, subjective information appears to matter for predicting survival.

6.2 Variable Importance: Leading Alternative Predictors of Longevity

In this section, I describe the top 10 most important features for predicting longevity in the random forest model and survival to 75 in the XGboost model.

Figure 5 displays the top 10 most important variables in the forest according to the permutation importance metric. The permutation performance metric is calculated by measuring how much the model's performance drops in terms of goodness of fit with the exclusion of a specific feature. Thus, highly important features that contribute the most to goodness of fit receive higher rankings. Unsurprisingly for predicting longevity, age deciles at different waves and the cumulative number of waves that an individual is apart of over time contribute the most to predicting longevity.

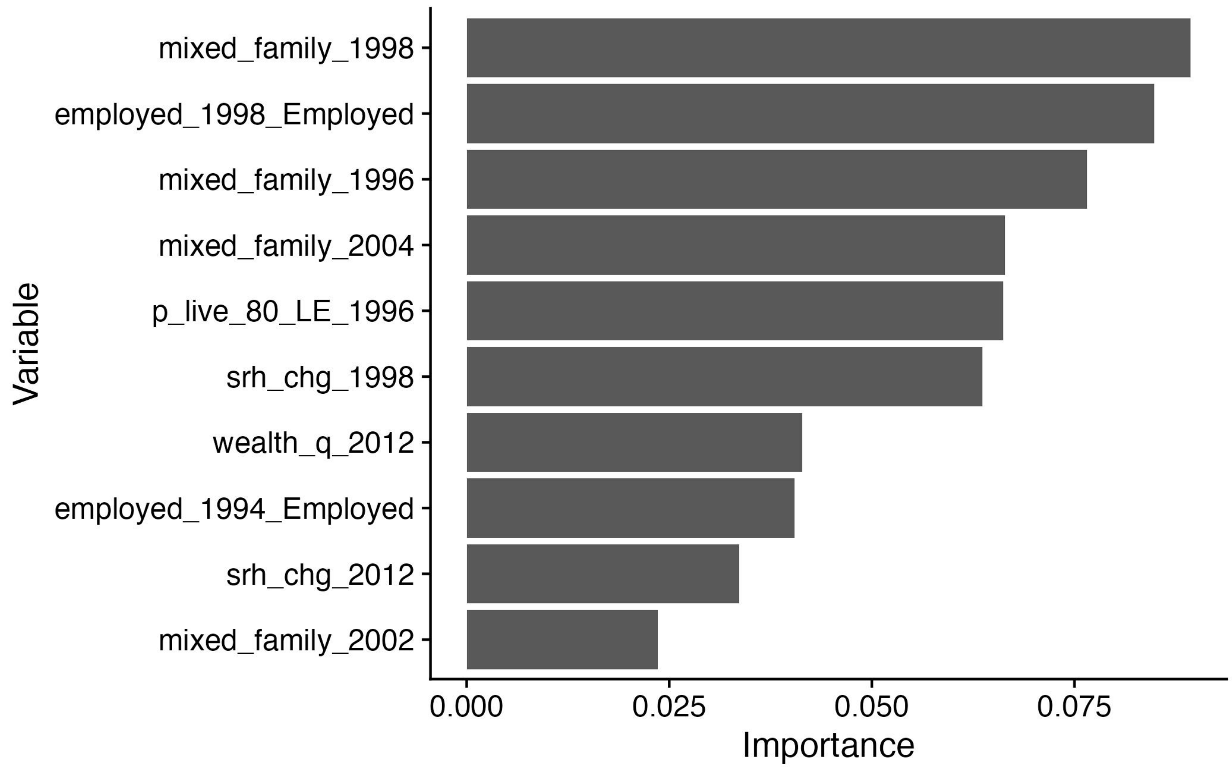


This plot describes the top 10 most relevant features using the permutation importance metric for my preferred random forest model with subjective probabilities and covariates. Higher values equal greater importance.

Figure 5: Top 10 Relevant Predictors of Longevity in Preferred Random Forest Model

Next Figure 6 displays the same feature importance metrics for the top 10 features in my XGBoost model predicting survival to age 75. Here the most important features are quite different. Mixed family status (as measured by whether the individual has biological and non-biological children) appear quite important.

Additionally, changes to self-rated health, employment, and wealth are predictive as well. "p_live_80_LE", the subjective probability that an individual survives past 80 years old also predicts survival to 75 strongly. This is unsurprising since individuals who have high subjective probabilities that they will survive past 80, let alone 75 years, may also have important unobserved information about wellbeing that is predictive of survival to 75.



This plot describes the top 10 most relevant features using the permutation importance metric for my preferred XGBoost model with subjective probabilities and covariates. Higher values equal greater importance.

Figure 6: Top 10 Relevant Predictors of Survival to Age 75 in Preferred XGBoost Model

7 Conclusion

Do subjective mortality expectations matter for predicting longevity? The answer is: it depends. Results from random forest models and several other machine learning methods suggest that subjective probabilities contribute little predictive information beyond what is obtained from other covariates. However, for predicting survival past the age of 75, the subjective mortality probabilities improve predictive power. These results highlight the potential utility of incorporating subjective measures into predictive models. Although subjective information provides weakly predictive information about longevity, it can improve the prediction of survival past thresholds.

8 Tables

Table 2: Summary of Model Performance (Longevity)

Model	Feature Set	OOS SD(Y)	OOS RMSE	OOS R^2	Overall Δ RMSE	Q1 Δ	Q2 Δ	Q3 Δ	Q4 Δ	Q5 Δ
OLS	Probs	9.283	9.116	0.036	0.000	0.000	0.000	0.000	0.000	0.000
OLS	Covariates	9.283	5.586	0.639	-38.724	-82.331	14.867	-156.775	-66.051	-46.377
OLS	Covariates + Probs	9.283	5.574	0.640	-38.851	-82.640	14.962	-157.432	-66.497	-46.545
LASSO	Covariates	9.283	5.581	0.639	-38.778	-82.720	12.584	-153.704	-64.983	-45.797
LASSO	Covariates + Probs	9.283	5.570	0.641	-38.902	-82.751	14.071	-156.674	-66.163	-46.388
Ridge	Covariates	9.283	5.568	0.640	-38.918	-80.599	10.447	-154.294	-65.089	-45.781
Ridge	Covariates + Probs	9.283	5.557	0.642	-39.036	-81.022	9.867	-154.081	-65.172	-45.768
Elastic	Covariates	9.283	5.580	0.639	-38.784	-82.350	14.399	-156.543	-65.904	-46.293
Elastic	Covariates + Probs	9.283	5.570	0.641	-38.903	-82.688	14.335	-157.023	-66.244	-46.413
Tree	Covariates	9.283	5.511	0.665	-39.541	-92.126	-1.616	-179.661	-81.966	-53.697
Tree	Covariates + Probs	9.283	5.486	0.668	-39.817	-90.083	-10.207	-182.129	-80.477	-56.425
Random Forest	Covariates	9.283	4.420	0.773	-51.514	-89.549	-1.222	-179.219	-79.761	-55.083
Random Forest	Covariates + Probs	9.283	4.445	0.771	-51.239	-89.551	-1.391	-178.694	-79.766	-54.647
XGBoost	Covariates	9.283	4.609	0.754	-49.445	-90.596	-5.614	-171.602	-79.473	-56.192
XGBoost	Covariates + Probs	9.283	4.606	0.754	-49.473	-90.417	-9.182	-173.570	-80.844	-56.820

Q columns refer to quantiles of improvement in RMSE over base OLS.

Table 3: Summary of Model Performance (Survival to age 75+)

Model	Feature Set	AUC	Accuracy	Sensitivity	Precision	Δ AUC	Δ Accuracy	Δ Sensitivity	Δ Precision
Logit	Probs	0.768	0.667	0.871	0.669	0.000	0.000	0.000	0.000
Logit	Covariates	0.834	0.744	0.799	0.777	8.606	11.588	-8.203	16.112
Logit	Covariates + Probs	0.885	0.805	0.842	0.833	15.217	20.811	-3.342	24.502
LASSO	Covariates	0.835	0.742	0.789	0.780	8.775	11.318	-9.375	16.600
LASSO	Covariates + Probs	0.885	0.805	0.842	0.832	15.174	20.709	-3.342	24.362
Ridge	Covariates	0.835	0.739	0.783	0.780	8.713	10.912	-10.113	16.609
Ridge	Covariates + Probs	0.867	0.781	0.827	0.810	12.856	17.162	-5.078	20.998
Elastic	Covariates	0.835	0.740	0.786	0.779	8.740	10.980	-9.722	16.414
Elastic	Covariates + Probs	0.884	0.805	0.842	0.832	15.155	20.709	-3.342	24.362
Tree	Covariates	0.887	0.797	0.854	0.815	15.436	19.628	-1.910	21.747
Tree	Covariates + Probs	0.917	0.844	0.889	0.855	19.435	26.622	2.040	27.800
Random Forest	Covariates	0.929	0.844	0.893	0.852	20.998	26.622	2.517	27.379
Random Forest	Covariates + Probs	0.953	0.878	0.907	0.890	24.137	31.655	4.123	32.973
XGBoost	Covariates	0.932	0.846	0.905	0.848	21.359	26.993	3.906	26.682
XGBoost	Covariates + Probs	0.964	0.893	0.915	0.907	25.477	34.054	5.122	35.505

Improvement is relative to a logistic regression with subjective probabilities as features

9 Appendix

Table 4: Tuning Paremeters by Model (Longevity)

Algorithm	Feature Set	N Predictors	Mixture	Lambda	M Try	Min Nodes	N Trees	Tree depth	Learn Rate	Sample size	CV Folds
OLS	-	647.00	-	-	-	-	-	-	-	-	-
Lasso	Controls	460.00	1	0	-	-	-	-	-	-	5
Lasso	Controls + Probs	483.00	1	0	-	-	-	-	-	-	5
Ridge	Controls	547.00	0	0	-	-	-	-	-	-	5
Ridge	Controls + Probs	547.00	0	0	-	-	-	-	-	-	5
Elastic Net	Controls	473.00	0.5	0	-	-	-	-	-	-	5
Elastic Net	Controls + Probs	487.00	0.75	0	-	-	-	-	-	-	5
Tree	Controls	647.00	-	-	435	40	1	-	-	-	5
Tree	Controls + Probs	647.00	-	-	435	40	1	-	-	-	5
Random Forest	Controls	647.00	-	-	145	30	100	-	-	-	5
Random Forest	Controls + Probs	647.00	-	-	151	30	100	-	-	-	5
XGBoost	Controls	647.00	-	-	-	40	100	8	0.3	1	5
XGBoost	Controls + Probs	647.00	-	-	-	40	100	8	0.3	1	5

All parameters are tuned using cross-validation except learn rate, number of trees, and number of folds.

10 References

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